Laboratory 1a
Rule-Based SOC Detection & MITRE ATT&CK Mapping

Course: AI-Enhanced Cybersecurity — Holon Institute of Technology (HIT)
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Platform: Pop!_OS Linux / Docker Engine 29.1.3 / Compose v2.40.3

1. Objective and architecture

The laboratory tests a deterministic rule-based detector using supplied and synthetic security logs. Matching rules produce candidate SOC findings enriched with local MITRE ATT&CK STIX metadata. No real network activity was scanned or attacked.

Input	Processing	Output
logs/security.log	Detector rescans every 5 seconds; local STIX maps technique IDs	output/findings.json and dashboard on localhost:8000

2. Docker verification and clean baseline

Service	Container	Status / port
detector	mitre-log-detector	Up
dashboard	lab1a-dashboard	Up / host port 8000

The clean security.log contained 6 records and was backed up before experiments. The initial detector output reported finding_count = 2, a failed-password threshold of 3, a 5-second rescan, and MITRE status Loaded.

Technique / tactic	Source IP	Severity	Evidence
T1110 / Credential Access	172.18.0.5	High	3 failed SSH password events
T1046 / Discovery	172.18.0.7	Medium	Explicit network-scan notice

3. Evidence → rule → technique → uncertainty

T1110 — Brute Force (Credential Access): three `Failed password` messages from source 172.18.0.5 triggered the rule requiring at least three matching lines for one source IP. These failed logins do not prove attacker intent or successful access.

T1046 — Network Service Discovery (Discovery): a line explicitly saying `Network scan detected` from 172.18.0.7 triggered the string-matching rule. It is not independent evidence from packets or network flows.

Figure 1 — Clean baseline dashboard

Verified screenshot: 2 total findings, including 1 High (T1110) and 1 Medium (T1046); source log contains the original six records.

Lab 1a SOC Dashboard

Cyclic detection with periodic MITRE ATT&CK STIX refresh

Last detector scan: 2026-11-04T17:59:29.882243+00:00
STIX status: Loaded
STIX source: enterprise-attack-mini.json
Techniques loaded: 2
STIX loaded at: 2026-11-04T17:47:24.533567+00:00

Total findings	High severity	Medium severity
2	1	1

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible brute-force login attempt	172.18.0.5	T1110 - Brute Force	Credential Access	High	Resolved from STIX	3 failed password events were detected from 172.18.0.5.

Raw Security Log

```
2026-07-11T09:00:10 INFO User student logged in successfully from 172.18.0.10
2026-07-11T09:02:11 WARN Failed password for root from 172.18.0.5 port 44231 ssh2
2026-07-11T09:02:14 WARN Failed password for root from 172.18.0.5 port 44232 ssh2
2026-07-11T09:02:18 WARN Failed password for root from 172.18.0.5 port 44233 ssh2
2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2026-07-11T09:07:33 INFO User administrator logged out
```

4. Controlled experiments C1–C4

The simulator was stopped during manual tests. After each edit, predictions were compared with actual findings.json outputs. All four predictions matched.

C1 — Below threshold

Changed log: Delete third failure from 172.18.0.5, port 44233. Two remain.
Predicted: 1; only T1046. Actual: 1; only T1046.
Explanation: Below threshold of 3.

C2 — Split source IPs

Changed log: Use 172.18.0.5, .6, .8 (one failure per IP).
Predicted: 1; only T1046. Actual: 1; only T1046.
Explanation: Per-IP grouping.

C3 — New-IP threshold

Changed log: Append 3 failures from 192.0.2.55.
Predicted: 2; T1046 + new T1110. Actual: 2; T1046 172.18.0.7 and T1110 192.0.2.55.
Explanation: New IP reaches threshold.

C4 — Second scan notice

Changed log: Append scan notice from 192.0.2.77.
Predicted: 3; T1046 x2 + T1110. Actual: 3; T1046 172.18.0.7 / 192.0.2.77 and T1110 192.0.2.55.
Explanation: String match adds finding.

Inserted synthetic log evidence

```
2026-10-22T10:01:00Z WARN Failed password for test from 192.0.2.55 port 40001 ssh2
2026-10-22T10:01:02Z WARN Failed password for test from 192.0.2.55 port 40002 ssh2
2026-10-22T10:01:04Z WARN Failed password for test from 192.0.2.55 port 40003 ssh2
2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,443
```

5. ATT&CK knowledge and simulator observation

The local enterprise-attack-mini.json loaded two techniques, both resolved from STIX: T1110 / Credential Access and T1046 / Discovery. In the simulator-stage screenshot, total findings rose from 2 to 3. The newly visible finding was T1046 from 172.18.0.31. New successful-login lines did not create rule matches at the captured moment.

This is a snapshot, not a claim that the count remained 3: the simulator appends new events over time, making its outcomes less reproducible than controlled manual edits.

6. SOC narrative (126 words)

On 4 November 2026, I tested a rule-based SOC detector in Docker on Pop!_OS. The clean six-line log produced two findings: T1110 (High) after three failed SSH password events from 172.18.0.5, and T1046 (Medium) for an explicit scan notice from 172.18.0.7. In C1, removing one failure eliminated T1110. In C2, distributing failures among three source IPs also avoided the threshold. In C3, three failures from 192.0.2.55 created a new T1110. In C4, a scan notice from 192.0.2.77 added T1046, giving three findings. A later simulator screenshot showed a new T1046 for 172.18.0.31. These observations confirm how the implemented rules classify log text; they do not establish hostile intent, a real scan, successful authentication by an attacker, or compromise. Corroboration would require independent endpoint and network evidence.

7. Detection limitations

The T1110 rule can miss low-volume or distributed authentication attempts and has no time window. T1046 trusts an explicit scan notice rather than observing packet-level activity. ATT&CK mapping does not establish malicious intent, attacker identity, successful access, or compromise.

Figure 2 — Simulator-stage dashboard

Observed screenshot: 3 findings (one High and two Medium), including the new network scan notice from 172.18.0.31.



8. Cleanup and reproducibility

After collecting evidence, restore the six-line baseline and stop all services. Do not commit experimental edits to the upstream security.log.

```
cp logs/security.log.bak logs/security.log
docker compose --profile simulation down
```

References

Lab source: github.com/melofon/HIT-ai-cybersecurity-labs/tree/main/labs/lab1a-mitre-soc
MITRE: attack.mitre.org/techniques/T1110/ & attack.mitre.org/techniques/T1046/